



មហាវិទ្យាល័យវិស្វកម្ម
FACULTY OF ENGINEERING

Computer Programing

Lecture 2 Getting Start with C++

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The history of C++

Year	Key Milestone
1989	Release of C++ 2.0 with multiple inheritance, abstract classes, static member functions, and const member functions.
1990	The inclusion of templates, enabling generic programming .
1998	ISO/IEC standardized C++ as C++98 , providing a stable, standardized version.
2003	C++03 , a bug-fix release to C++98 with no major new features.
2011	C++11 (C++0x) introduced major modern features: auto, nullptr, lambda expressions, smart pointers, move semantics, range-based for loops, and concurrency support.
2014	C++14 , a small extension over C++11 with bug fixes and minor improvements.
2017	C++17 , included features like structured bindings, if constexpr, and improved standard library components.
2020	C++20 , a major update with concepts, ranges, coroutines, modules, and more powerful compile-time programming.
Upcoming	C++23 and further versions continue to refine and expand the language for performance and software engineering needs.

Why Learn C++?

- C++ is used to **develop games, desktop apps, operating systems, browsers**, and so on because of its performance.
- After learning C++, it will be much easier to learn other programming languages like **Java, Python**, etc.
- C++ helps you to understand the internal architecture of a computer, and how **computer stores and retrieves information**.

How to learn C++?

- Interactive C++ Course
- C++ tutorial from any source
- Official C++ documentation
- Write a lot of C++ programming code
- Read C++ code

Basic Structure of a C++ program

Describe the C++ program below

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Hello World!" << endl;
6      return 0;
7  }
```

Basic Structure of a C++ program

Describe the C++ program below

- **Standard libraries section**
 - **#include <iostream>** – for cin, cout, cerr

```
1  #include <iostream>
2  using namespace std;
```

- Main function section
- Function body section

Basic Structure of a C++ program

Describe the C++ program below

- Standard libraries section
- **Main function section: write your code**

```
4 ✓ int main() {  
5     cout << "Hello World!" << endl;  
6     return 0;  
7 }
```

- Function body section

Basic Structure of a C++ program

Describe the C++ program below

- Standard libraries section
- **Main function section**

Use-case	return 0	return 1
In the main function	return 0 in the main function means that the program executed successfully.	return 1 in the main function means that the program does not execute successfully and there is some error.

- Function body section

#include <iostream> in C++

- Standard *input-output* stream
- Declares objects that control reading from and writing to the standard streams
- Provides input and output functionality using streams
- Syntax: `#include <iostream>`

What is #include <iostream> in C++?

1. Standard Output Stream -- **cout**

=> It produces output on the standard output device, i.e., the display screen.

```
#include <iostream>
using namespace std;
int main()
{
    cout << "My name is Tida" << endl;
    cout << 39 << endl;
    cout << true << endl;
    cout << 56.8 << endl;
    cout << 'A' << endl;
    return 0;
}
```

```
My name is Tida
39
1
56.8
A
```

#include <iostream> in C++

2. Standard Input Stream – cin

=> It reads input from the standard input device, i.e., the keyboard.

```
#include <iostream>
using namespace std;
int main() {
    int age;
    cout << "Enter your age: ";
    cin >> age;
    cout << "You entered: " << age << endl;
    return 0;
}
```

Enter your age: 19
You entered: 19

#include <iostream> in C++

```
#include <iostream>
using namespace std;

int main() {
    int a, b, sum;

    cout << "Enter first number: ";
    // Read input from the standard input device, usually the keyboard.
    cin >> a;

    cout << "Enter second number: ";
    // Read another input from the keyboard.
    cin >> b;

    sum = a + b;

    cout << "The sum of the two numbers is " << sum;

    return 0;
}
```

```
Enter first number: 21
Enter second number: 35
The sum of the two numbers is 56
```

using namespace std

- A **namespace** is a way to group names (like classes, functions, and variables) to avoid naming conflicts.
- For example, in the Standard Library (std), you have functions like **cout**, **cin**, **string**, etc.

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World" << endl;
    return 0;
}
```

```
#include <iostream>
int main() {
    std::cout << "Hello World" << std::endl;
    return 0;
}
```

escape sequences

- **escape sequences** are used to represent characters that are difficult or impossible to express directly in a string or character literal

Escape Sequence	Meaning
<code>\n</code>	Newline
<code>\t</code>	Horizontal tab

escape sequences

```
1  #include <iostream>
2  using namespace std;
3  ✓ int main()
4  {
5      cout << "Name\t\tAge\tCity\n";
6      cout << "Tida\t\t30\tSiem Reap\n";
7      cout << "Dara\t\t22\tPhnom Penh\n";
8
9      return 0;
10 }
```

Name	Age	City
Tida	30	Siem Reap
Dara	22	Phnom Penh

CP – W2 – Practice

Ex 1

Write a program in C++ to print a welcome text in a separate line.

Expected output:

```
Print a welcome text in a separate line :
```

```
First time to practice C++
```

```
FE, Telecom & Electronic Department
```

```
Computer programming with C++
```

Ex 2

Write a C++ program that

Expected output:

```
My friend Infomation :
```

```
-----  
My friend Name :           Tida  
The Gender is :           F  
Date of Birth :           2009  
Place of Birth is :       Siem Reap  
GPA is                    4.69
```

Ex 3

Write a C++ program that defines two variables for floating-point numbers and initializes them with the values

Expected output:

```
Total: 123.456 + 76.543 = 199.999
```

```
Difference: 123.456 - 76.543 = 46.913
```

Ex 4

Write a program in C++ to add two numbers accept through keyboard

Expected output:

```
Sum of two numbers :
```

```
-----
```

```
Input 1st number : 25
```

```
Input 2nd number : 39
```

```
The sum of the numbers is : 64
```

Thanks!